COUNTRY CLUSTERING & ANALYSIS

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Tools used:

Python 3

Jupyter Notebook

Libraries used:

Pandas (Data manipulation)

Numpy (Numerical functions)

Sklearn (Machine Learning algorithms)

Seaborn (Visualization)

Plotly (Visualization)

Scipy (Statistics functions)

Introduction

In the world of intelligence, in terms of operating a business, optimizing a process, uncovering various elements that boost workforce performance etc., the most valuable resource is data. In this project, I tried to create something valuable, interpretable, and meaningful using various country data from 2023. Turning data into useful insights consists of steps such as cleaning the data, transforming values, analyzing and visualizing relationships between features, training a machine learning model, and interpreting the results... In this project, my purpose was not to find the countries with the biggest army size, biggest GDP, largest land area, most crowded population, etc. My purpose was to go deeper into people's lives and create segments that will show us people's overall quality of life around the world. Thus, I have to warn that this dataset is not specifically designed around my purpose and does not include many metrics that point out the quality of life in that country. This project is a passion project that intends to give me the opportunity to put what I've learned into practice about the field of data analytics. Thank you in advance for reading my project.

Understanding the Data

	Country	Density\n(P/Km2)	Abbreviation	Agricultural Land(%)	Land Area(Km2)	Armed Forces size	Birth Rate	Calling Code	Capital/Major City	Co2-Emissions	CPI	CPI Change (%)	Currency-Code	Fertility Rate	Forested Area (%)
0	Afghanistan	60	AF	58.10%	652,230	323,000	32.49	93.0	Kabul	8,672	149.9	2.30%	AFN	4.47	2.10%
1	Albania	105	AL	43.10%	28,748	9,000	11.78	355.0	Tirana	4,536	119.05	1.40%	ALL	1.62	28.10%
2	Algeria	18	DZ	17.40%	2,381,741	317,000	24.28	213.0	Algiers	150,006	151.36	2.00%	DZD	3.02	0.80%
3	Andorra	164	AD	40.00%	468	NaN	7.20	376.0	Andorra la Vella	469	NaN	NaN	EUR	1.27	34.00%
4	Angola	26	AO	47.50%	1,246,700	117,000	40.73	244.0	Luanda	34,693	261.73	17.10%	AOA	5.52	46.30%
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
190	Venezuela	32	VE	24.50%	912,050	343,000	17.88	58.0	Caracas	164,175	2,740.27	254.90%	VED	2.27	52.70%
191	Vietnam	314	VN	39.30%	331,210	522,000	16.75	84.0	Hanoi	192,668	163.52	2.80%	VND	2.05	48.10%
192	Yemen	56	YE	44.60%	527,968	40,000	30.45	967.0	Sanaa	10,609	157.58	8.10%	YER	3.79	1.00%
193	Zambia	25	ZM	32.10%	752,618	16,000	36.19	260.0	Lusaka	5,141	212.31	9.20%	ZMW	4.63	65.20%
194	Zimbabwe	38	ZW	41.90%	390,757	51,000	30.68	263.0	Harare	10,983	105.51	0.90%	NaN	3.62	35.50%

(Original csv of the dataset has been provided. You can look at the dataset by yourself.)

This dataset has 35 columns and 195 rows. Thus, 195 unique countries and 35 features that describe them. As you can see from the first 5 and last 5 country names sorted alphabetically, we have null values (NaN) and numerical values with a string symbol (%). As depicted, this dataset is compiled from multiple data sources.

Diagnosis

```
print("--- Data Info ---")
df.info()

print(f"\n--- Null Values ---\n{df.isnull().sum()}")

print(f"\n--- Duplicate Values ---\n{df.duplicated().sum()}")

print("--- Stats of Data ---")
print(df.describe())

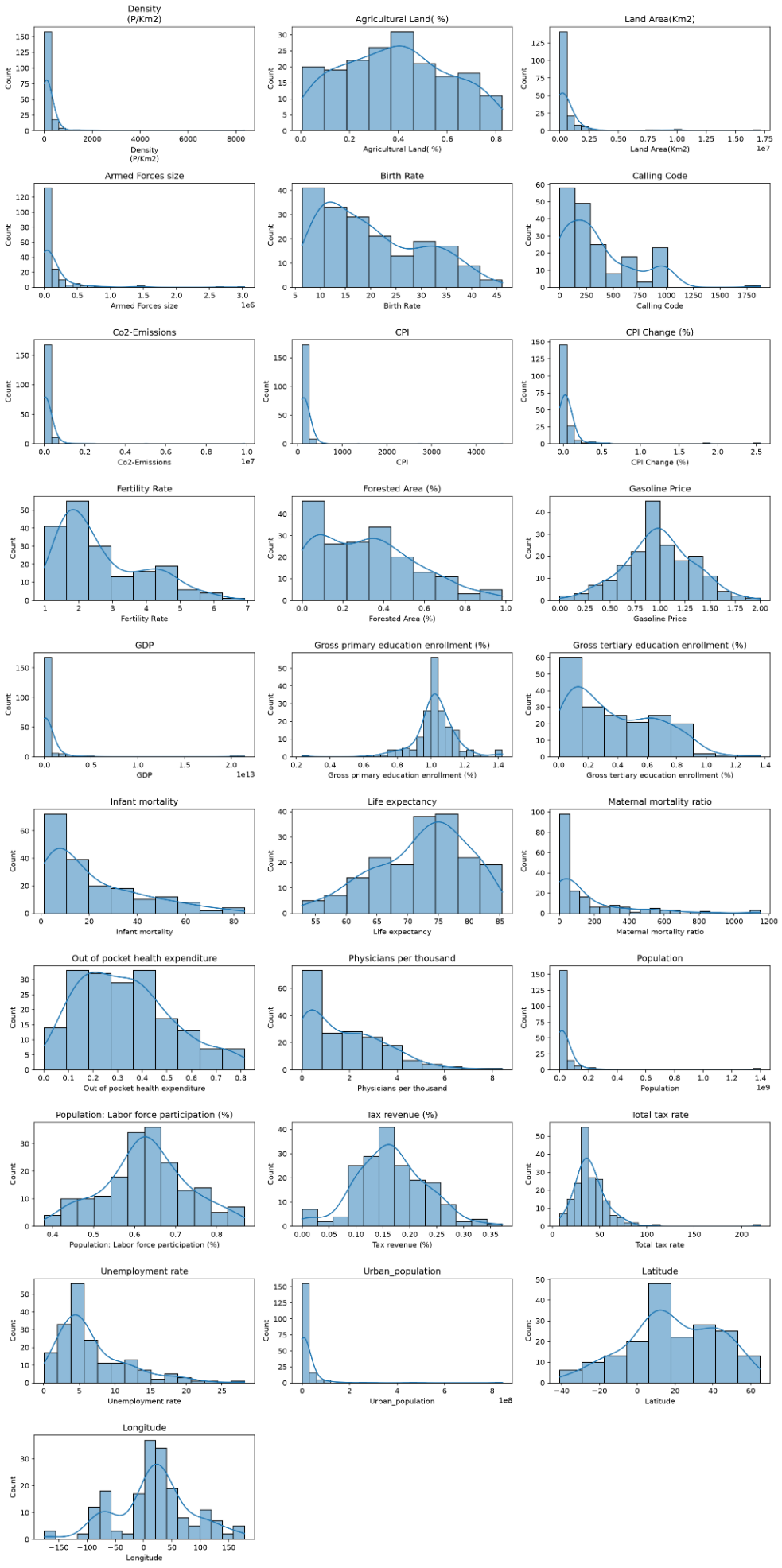
missing_per_row = df.isna().sum(axis=1)
missing_summary = pd.DataFrame({'Country': df['Country'], 'Missing_Count': missing_per_row}).sort_values('Missing_Count', ascending=False)
print("\n--- Countries With Lots of Null Values ---")
print(missing_summary.head(20))
```

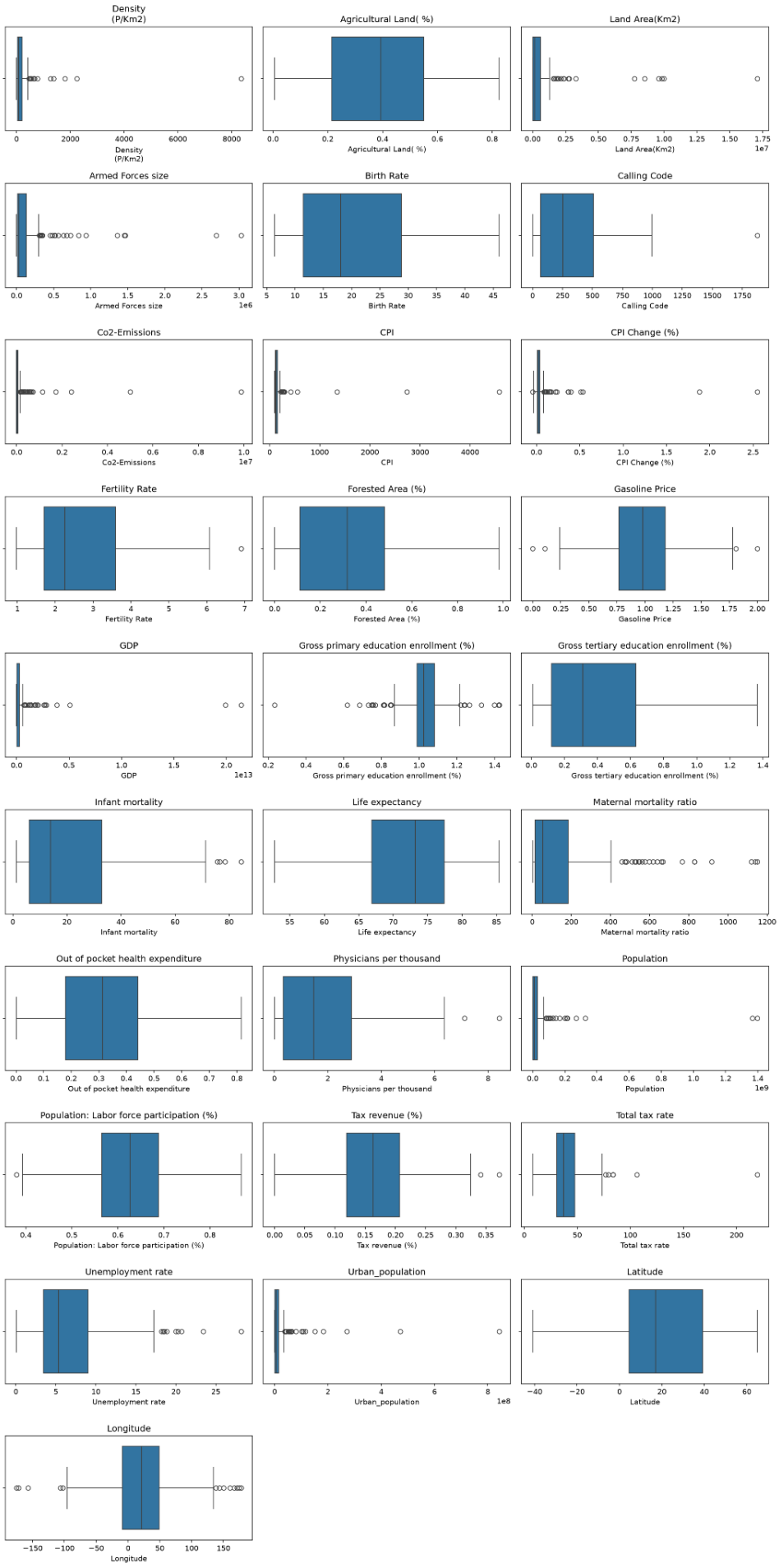
When I look at the information, I see lots of data type mismatches. Most of the columns have been assigned the 'string' data type because of the symbols they contain, such as percent and dollar signs. Also, there are columns that were assigned the 'integer' data type even though they contain floating-point values.

Out of 35 columns, there are null values in 33 of them. The minimum wage column has the most null values with a count of 45, which is normal because of different minimum wage policies around the world. The overall amount of null values is normal for a dataset of this size.
On the other hand, there are various countries that have a lot of missing values. For example, Palestine has 30, Vatican City has 25, Nauru has 23, North Macedonia has 22, Eswatini has 22, etc. The reason for this could be related to political matters or simply data entry issues.

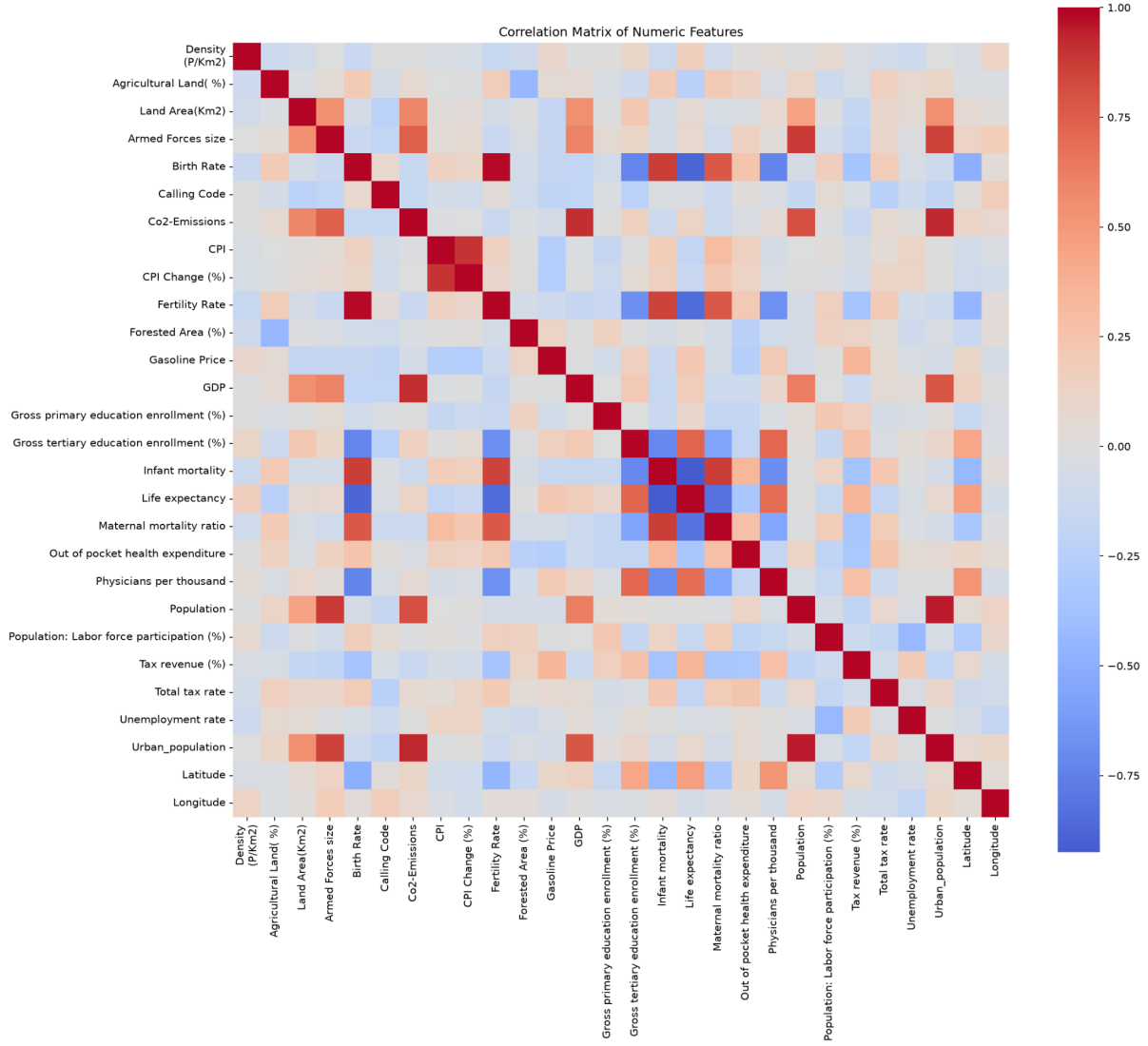
```
--- Countries With Lots of Null Values ---
                            Country  Missing_Count
133  Palestinian National Authority             30
73                      Vatican City             25
120                            Nauru             23
128                  North Macedonia             22
56                          Eswatini             22
113                           Monaco             15
181                           Tuvalu             14
98                     Liechtenstein             11
3                             Andorra             10
```

Then I looked at the distribution of numerical columns using histogram and boxplot. It is important to detect the overall behavior of the data and outliers.


Density (P/Km2)
Agricultural Land(%)
Land Area(Km2)
Armed Forces size
Birth Rate
Calling Code
Co2-Emissions
CPI
CPI Change (%)
Fertility Rate
Forested Area (%)
Gasoline Price
GDP
Gross primary education enrollment (%)
Gross tertiary education enrollment (%)
Infant mortality
Life expectancy
Maternal mortality ratio
Out of pocket health expenditure
Physicians per thousand
Population
Population: Labor force participation (%)
Tax revenue (%)
Total tax rate
Unemployment rate
Urban_population
Latitude
Longitude
Count


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As we can observe, there are features that have a normal distribution, features that are tilted to the right (right-skewed), and features that are tilted to the left (left-skewed). The points outside the box plots consist of outlier observations, the same outliers that skew our data to the right or left. In a proper dataset to analyze and implement machine-learning algorithms, we had to handle the outliers and the overall skewness. Countries that cause these outliers mostly consist of very major countries such as the USA, China, India, Russia, Japan. Or some countries that have enormous focus on various elements such as army size per thousand people.



Then, I used a heatmap to see the correlation between features. As we know, correlation can be negative or positive. For example, when a socio-economic indicator

of a neighborhood increases, crime decreases, which points out a negative correlation. Or, when the socio-economic indicator of a neighborhood increases, the satisfaction level increases, which points out a positive correlation.

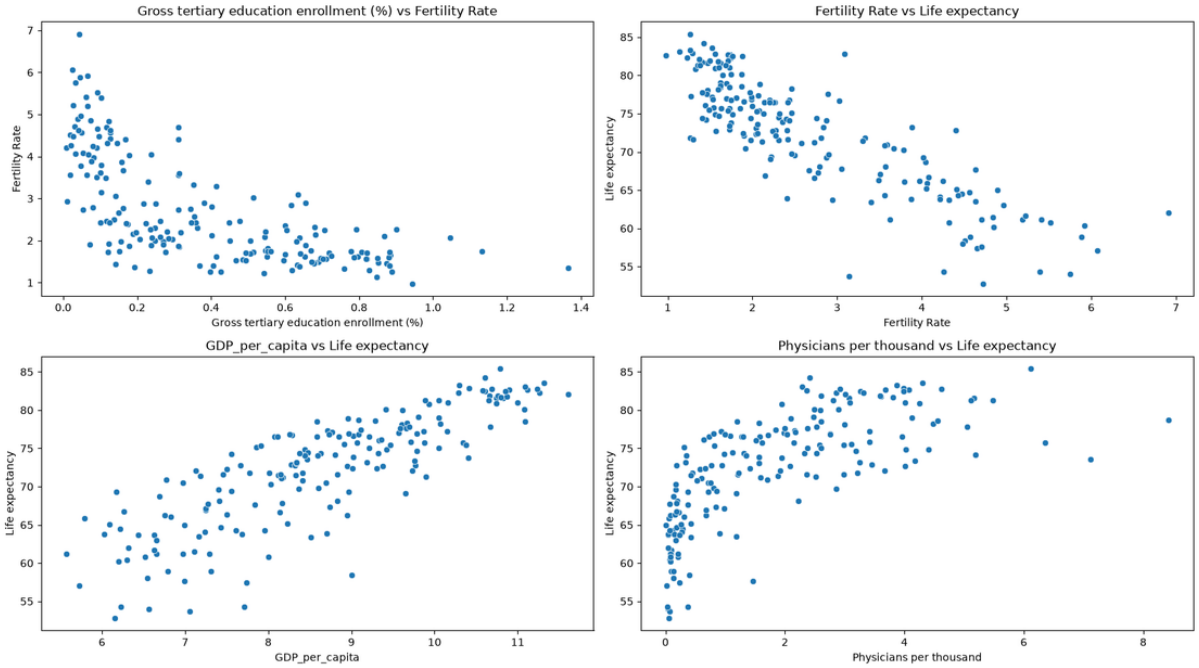
But there is one key element to this: collinearity. When two features have a very strong correlation with each other, that might be a sign of collinearity. For example, we have two features: the square footage of apartments and the number of rooms. A high number of rooms obviously points to more square footage, thus we don't need both of them. If you include both of them to train your machine learning algorithm, it can cause problems such as the curse of dimensionality, loss of interpretability, and broken regression coefficients. So, we will drop some columns on the cleaning phase according to our correlation matrix.

Cleaning

- *Percentage and dollar symbols are removed and relevant columns transformed into correct data types.*
- *Other columns are transformed into correct data types.*
- *Removed the countries with null columns more than 10.*
- *Null values on columns that would be pointless to impute have been dropped.*
- *Other columns are imputed with the median value of that column, because of the possible outliers.*
- *Removed the minimum wage column as a whole.*
- *Removed the 'Birth Rate', 'Infant Mortality' and 'Maternal Mortality 'Ratio' columns because they were highly positive/negative correlated with Fertility Rate and Life Expectancy.*
- *Created a 'GDP Per Capita' column using the formula GDP / Population and took the logarithm to ensure normal distribution.*

Clustering

After the cleaning data we are ready to prepare our data for clustering. In this stage we have to choose features that will define our clusters. I pick Gross Tertiary Education Enrollment, Fertility Rate, Life Expectancy, GDP Per Capita and Physicans per Thousand. Among five of them, only the fertility rate might look strange among them. I choose that feature because there is a strict pattern and negative correlation between a country's development and fertility rate.



	Fertility Rate	Life expectancy	GDP_per_capita	Gross tertiary education enrollment (%)	Physicians per thousand
count	185.000000	185.000000	185.000000	185.000000	185.000000
mean	2.703892	72.232973	8.650053	0.379627	1.818865
std	1.279453	7.480885	1.448462	0.290618	1.651716
min	0.980000	52.800000	5.569289	0.008000	0.010000
25%	1.710000	66.900000	7.505243	0.123000	0.340000
50%	2.250000	73.200000	8.692173	0.312000	1.460000
75%	3.590000	77.400000	9.757158	0.631000	2.890000
max	6.910000	85.400000	11.609811	1.366000	8.420000

Preparing

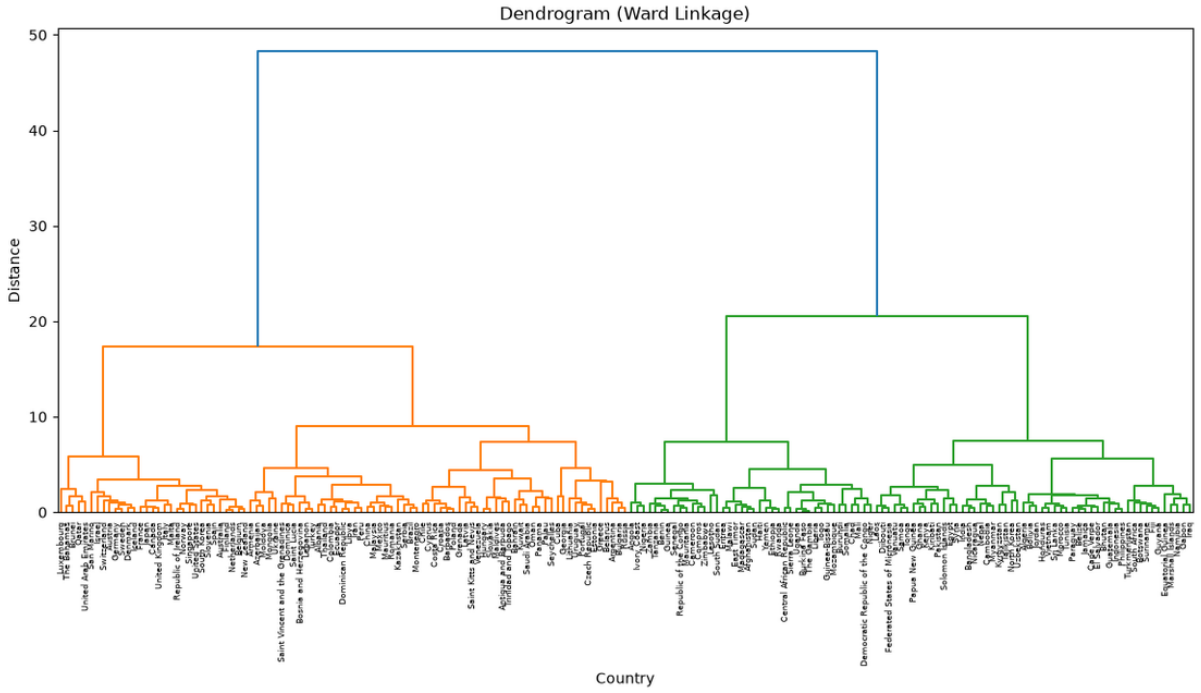
In order to make use of our features we have to standardize them. Standardization ensures that large numbers (mostly outliers) don't affect our clustering process unnaturally. The StandardScaler method uses the Z-Score formula (z = original value - mean of feature / standard deviation of the future) so that each feature has a mean of 0 and standard deviation of 1.

At this stage, I created a new DataFrame named 'df_scaled' with the features I picked. Also multiplied the GDP Per Capita column by 2.5, because I don't have any other economic metric than this. This was a personal choice of mine and I think I got better results from it.

	Fertility Rate	Life expectancy	GDP_per_capita	Gross tertiary education enrollment (%)	Physicians per thousand
count	1.850000e+02	1.850000e+02	1.850000e+02	1.850000e+02	1.850000e+02
mean	-4.680940e-17	1.773956e-15	-8.257659e-16	9.121832e-17	2.880579e-17
std	1.002714e+00	1.002714e+00	2.506784e+00	1.002714e+00	1.002714e+00
min	-1.351022e+00	-2.604733e+00	-5.331732e+00	-1.282219e+00	-1.098115e+00
25%	-7.789178e-01	-7.148145e-01	-1.981267e+00	-8.854366e-01	-8.977805e-01
50%	-3.557172e-01	1.296172e-01	7.289538e-02	-2.333326e-01	-2.178576e-01
75%	6.944471e-01	6.925717e-01	1.916015e+00	8.673086e-01	6.502584e-01
max	3.296347e+00	1.764866e+00	5.122312e+00	3.403269e+00	4.007378e+00

(Our features after scaling.)

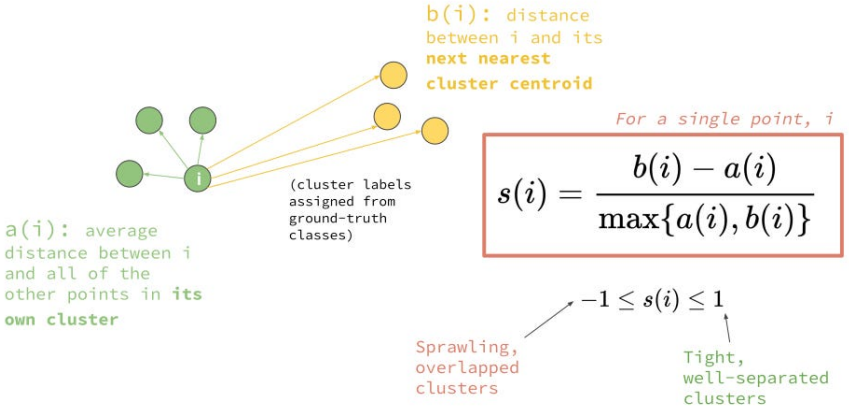
After the scaling process, we have to define the amount of clusters we will have. We will use the ‘Ward’ method. Ward Method is a criterion used in hierarchical clustering that minimizes the total within-cluster variance when clusters are combined to create similar clusters.



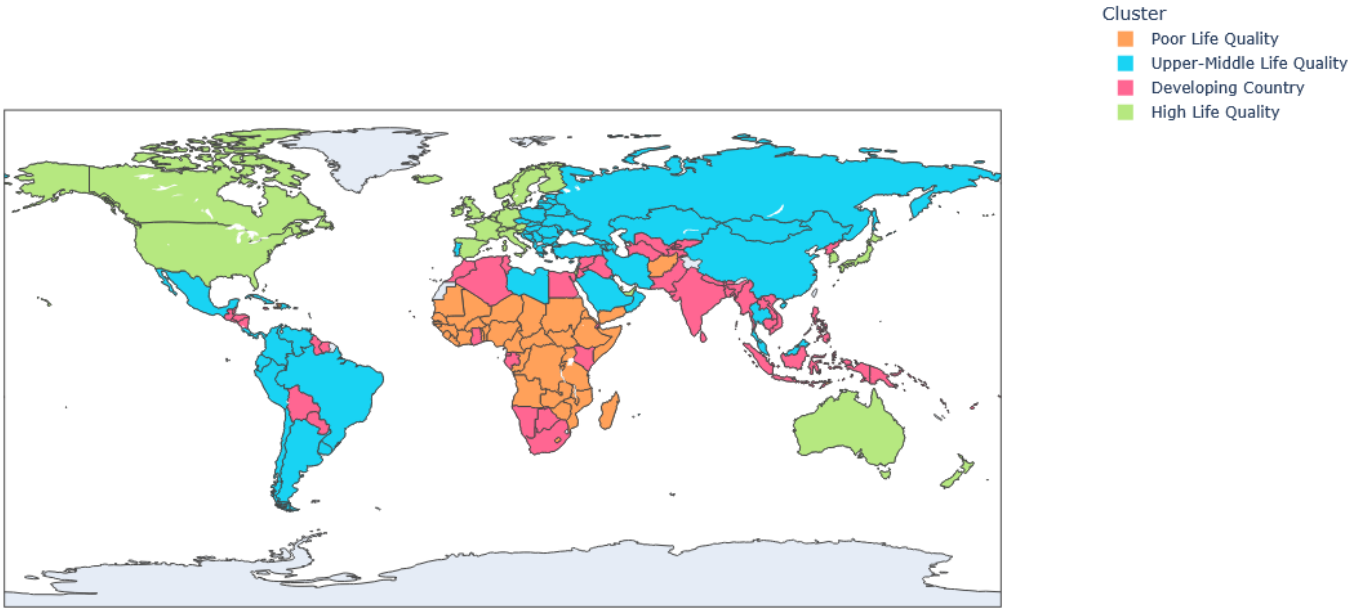
I used the dendrogram graph to see how the 'Ward' method separated our countries. As we can see it starts at the bottom with lots of differentiated clusters and ends with the one final cluster (blue). We have to pick a suitable amount of clusters. But dendrogramming is not enough. So, we can benefit from the silhouette score.

k	silhouette_score
2	0.483943
3	0.382401
4	0.346865
5	0.292088
6	0.249243
7	0.232407
8	0.236482
9	0.242463
10	0.240606

Silhoutte score is a metric to evaluate the quality of clustering models.



Interpretation




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--- Cluster Developing Country ---

['Algeria', 'Bangladesh', 'Belize', 'Bhutan', 'Bolivia', 'Botswana', 'Cape Verde', 'Cambodia', 'Djibouti', 'Egypt', 'El Salvador', 'Equatorial Guinea', '
Fiji', 'Gabon', 'Ghana', 'Guatemala', 'Guyana', 'Honduras', 'India', 'Indonesia', 'Iraq', 'Jamaica', 'Jordan', 'Kenya', 'Kiribati', 'Kyrgyzstan', 'Laos',
'Marshall Islands', 'Federated States of Micronesia', 'Morocco', 'Myanmar', 'Namibia', 'Nepal', 'Nicaragua', 'North Korea', 'Pakistan', 'Papua New Guinea
', 'Paraguay', 'Philippines', 'Samoa', 'Solomon Islands', 'South Africa', 'Sri Lanka', 'Suriname', 'Syria', 'Tajikistan', 'Tonga', 'Tunisia', 'Turkmenist
an', 'Uzbekistan', 'Vanuatu', 'Vietnam']

--- Cluster High Life Quality ---

['Australia', 'Austria', 'The Bahamas', 'Belgium', 'Brunei', 'Canada', 'Denmark', 'Finland', 'France', 'Germany', 'Iceland', 'Republic of Ireland', 'Isra
el', 'Italy', 'Japan', 'Luxembourg', 'Malta', 'Netherlands', 'New Zealand', 'Norway', 'Qatar', 'San Marino', 'Singapore', 'Slovenia', 'South Korea', 'Spa
in', 'Sweden', 'Switzerland', 'United Arab Emirates', 'United Kingdom', 'United States']

--- Cluster Poor Life Quality ---

['Afghanistan', 'Angola', 'Benin', 'Burkina Faso', 'Burundi', 'Ivory Coast', 'Cameroon', 'Central African Republic', 'Chad', 'Comoros', 'Republic of the
Congo', 'Democratic Republic of the Congo', 'Eritrea', 'Ethiopia', 'The Gambia', 'Guinea', 'Guinea-Bissau', 'Haiti', 'Lesotho', 'Liberia', 'Madagascar',
'Malawi', 'Mali', 'Mauritania', 'Mozambique', 'Niger', 'Nigeria', 'Rwanda', 'Senegal', 'Sierra Leone', 'Somalia', 'South Sudan', 'Sudan', 'Tanzania', 'Ea
st Timor', 'Togo', 'Uganda', 'Yemen', 'Zambia', 'Zimbabwe']

--- Cluster Upper-Middle Life Quality ---

['Albania', 'Antigua and Barbuda', 'Argentina', 'Armenia', 'Azerbaijan', 'Bahrain', 'Barbados', 'Belarus', 'Bosnia and Herzegovina', 'Brazil', 'Bulgaria'
, 'Chile', 'China', 'Colombia', 'Costa Rica', 'Croatia', 'Cuba', 'Cyprus', 'Czech Republic', 'Dominica', 'Dominican Republic', 'Ecuador', 'Estonia', 'Geo
rgia', 'Greece', 'Grenada', 'Hungary', 'Iran', 'Kazakhstan', 'Kuwait', 'Latvia', 'Lebanon', 'Libya', 'Lithuania', 'Malaysia', 'Maldives', 'Mauritius', 'M
exico', 'Moldova', 'Mongolia', 'Montenegro', 'Oman', 'Palau', 'Panama', 'Peru', 'Poland', 'Portugal', 'Romania', 'Russia', 'Saint Kitts and Nevis', 'Sain
t Lucia', 'Saint Vincent and the Grenadines', 'Saudi Arabia', 'Serbia', 'Seychelles', 'Slovakia', 'Thailand', 'Trinidad and Tobago', 'Turkey', 'Ukraine',
'Uruguay', 'Venezuela']
```

When looking at the silhouette score, I optimally should have chosen two, or perhaps three clusters. However, for the sake of this project, I chose four clusters because it will be easier to interpret. Countries vary a lot, and I think three clusters don't explain the differences in quality of life very well. With a few exceptions I liked the results I got.

Cluster	Fertility Rate	Life expectancy	GDP_per_capita	Gross tertiary education enrollment (%)	Physicians per thousand
Developing Country	2.857308	70.473077	8.066371	0.216462	0.875769
High Life Quality	1.594839	81.561290	10.786919	0.670032	3.362903
Poor Life Quality	4.675250	61.630000	6.715542	0.081825	0.203000
Upper-Middle Life Quality	1.857903	75.885484	9.319230	0.563403	2.880323

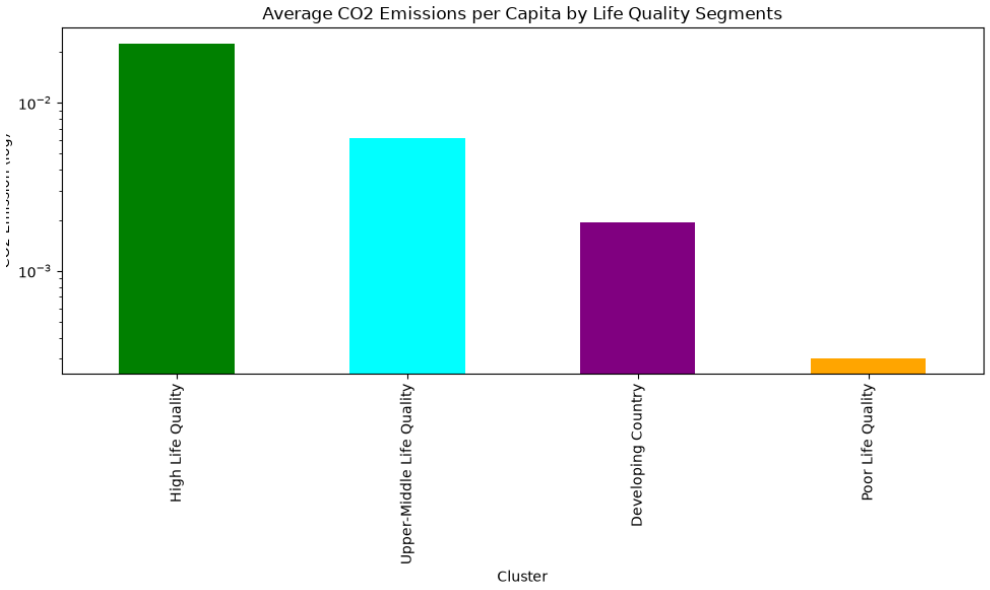
Countries within the High Life Quality segment have significantly lower fertility rates, a very long life expectancy, higher GDP per capita, higher GTEE, and more physicians per thousand people. These are the countries that have a very strong purchasing power, a long life expectancy, and a highly educated society with very low fertility rates. It shows that the richer a country is, the lower the fertility rate. I think the low life expectancy of countries within the Poor Life Quality segment can be explained by an insufficient number of physicians and many children. It is a sad reality.

And we also have the Upper-Middle Life Quality segment, which consists of countries very similar to those in the High Life Quality segment. They have a very similar fertility rate, a life expectancy that is six points lower, a very close GDP per capita, and lower GTEE and physicians per thousand. Since the High Life Quality cluster includes countries that industrialized almost 300 years ago, it is very hard to catch up, but the countries within this cluster are almost making it.

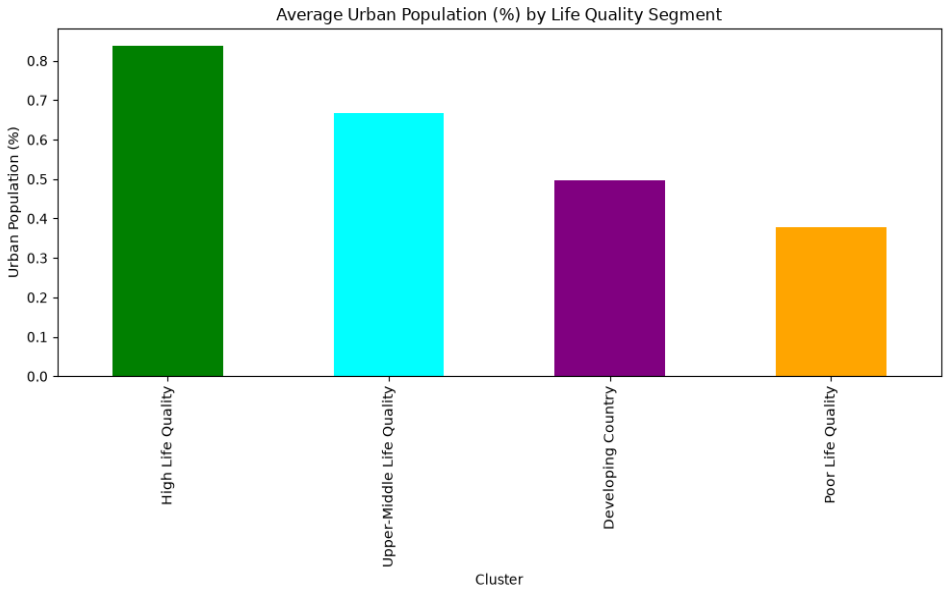
Lastly, we have the Developing Country cluster. This cluster is by far superior to the countries within the Poor Life Quality segment, but the difference can be better explained by comparison with the Upper-Middle Life Quality segment. It has a higher fertility rate, a life expectancy that is five years lower, a similar GDP per capita, and significantly lower GTEE and physicians per thousand. This segment mostly consists of relatively small Asian countries. And that makes sense. The market is shifting

towards Asia. They have begun to develop their economies, thus in the near future they will have a more educated society and more physicians.

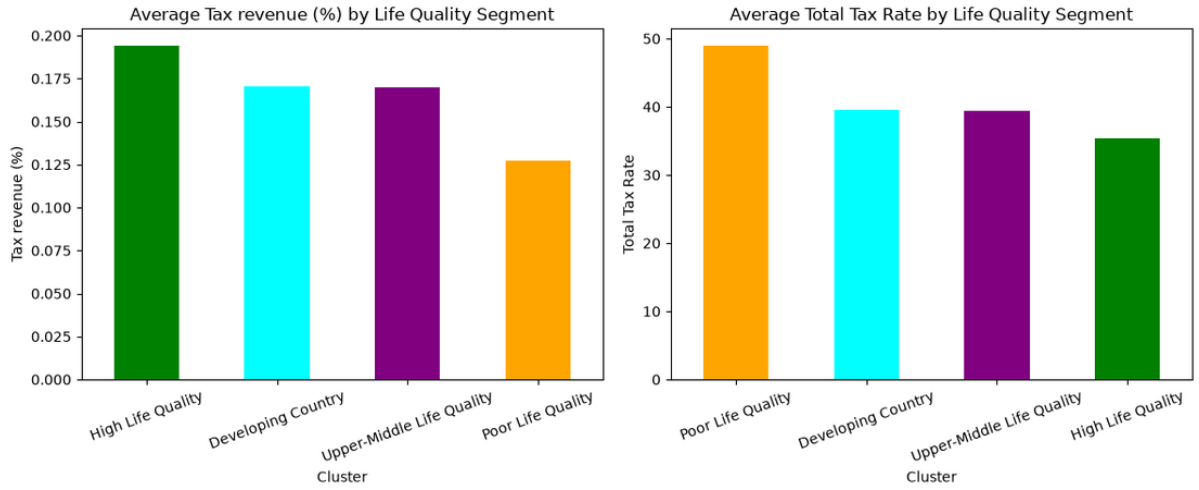
Findings



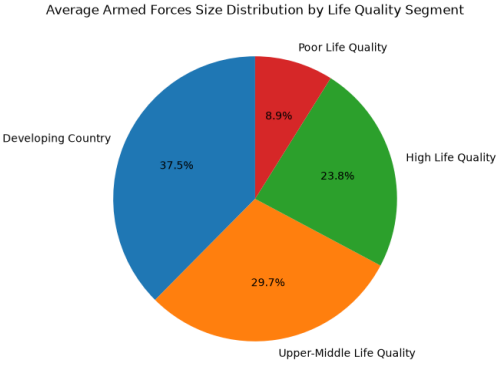
Our data shows that life quality and CO2 emission has positive correlation. It is expected because the countries with high life quality are also very industralized. There are lots of factories, lots of production...



Our data also shows that urban population rate also correlates with life quality. Most countries with poor life quality still have a huge population in urban areas unlike the other segments.



And the visual above shows us an interesting insight. While the average tax revenue was higher in the countries with high life quality, the average total tax rate was significantly higher in the countries with poor life quality. I think that shows us the state's capacity to efficiently collect tax money.



And lastly, it seems that the average armed forces size distribution is fairly equal across all clusters, except for the countries with poor life quality. In comparison with developing and upper-middle countries, countries with high life quality do not show superiority here.